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#Date: 2026-09-12
#By: Divyam Sood
#Leap Year

.data
input: .asciiz "Enter year:"
yes: .asciiz "Input year is a leap year"
no: .asciiz "Input year is not a leap year"



.globl main
.text

main:
    li $v0, 4
    la $a0, input
    syscall
    li $v0, 5
    syscall
    move $s0, $v0
    andi $t1, $s0, 3 #and to check division by power of 2
    bne $t1, $zero, notleap
    #This means year is divisible by 4. now we check divisible by 100.
    li $t0, 100
    div $s0, $t0
    mfhi $t1
    bne $t1, $zero, leap #divisible by 4 and not by 100
    #Now divisble by 4 and 100. so should be divisble by 400
    li $t0, 400
    div $s0, $t0
    mfhi $t1
    beq $t1, $zero, leap
    bne $t1, $zero, notleap


notleap:
    la $a0, no
    li $v0, 4
    syscall
    beq $zero, $zero, end

leap:
    la $a0, yes
    li $v0, 4
    syscall
    beq $zero, $zero, end



end:
    # exit program
    li  $v0, 10         # system call code for exit = 10
    syscall             # call operating system
```